

Tetris In Java Swing (With UML Implementation) - Use Cases

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1. Startup

Description: System starts up and starts the user at the main menu

Pre-condition: The system is not running

Post-condition: The system starts up and the user is at the main menu or the program can't run

Use Case 1.1: Successful Startup

1. User starts the system
2. System starts up with no issues and takes the user to the main menu
3. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 1.2: Unsuccessful Startup

1. User starts the system
2. System encounters an error at startup and can't take the user to the main menu
3. END

Next Use Case: Startup

2. Classic Mode

Description: User selects to play the Classic Mode from the main menu.

Pre-condition: User selects to play the Classic Mode from the main menu.

Post-condition: User gets a game over and views their score and can choose to play again or exit to the main menu.

Use Case 2.1: Soft Drop

1. The game begins with a random tetrominos falling.
2. The user lets the game automatically drop a piece.
3. The game awards the user 18 points (18 cells the piece moves)
4. The user gets a game over.
5. The program displays the user their score.
6. The program asks if the user wants to play again or exit to the main menu.
7. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.2: Hard Drop

1. The game begins with a random tetrominos falling.
2. The user lets the game automatically drop a piece.
3. The game awards the user 36 points (18 cells the piece moves)
4. The user gets a game over.
5. The program displays the user their score.
6. The program asks if the user wants to play again or exit to the main menu.
7. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.3: Single Line Clear

1. The game begins with a random tetrominos falling.
2. The user creates a stack of tetrominos on the grid.
3. The user uses a tetrominos to clear a line.
4. The game rewards the user 100 points times the level of the game.
5. The user gets a game over.
6. The program displays the user their score.
7. The program asks if the user wants to play again or exit to the main menu.
8. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.4: Double Line Clear

1. The game begins with a random tetrominos falling.
2. The user creates a stack of tetrominos on the grid.
3. The user uses a tetrominos to clear two lines.
4. The game rewards the user 300 points times the level of the game.
5. The user gets a game over.
6. The program displays the user their score.
7. The program asks if the user wants to play again or exit to the main menu.
8. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.5: Triple Line Clear

1. The game begins with a random tetrominos falling.
2. The user creates a stack of tetrominos on the grid.
3. The user uses a tetrominos to clear three lines.
4. The game rewards the user 500 points times the level of the game.
5. The user gets a game over.
6. The program displays the user their score.
7. The program asks if the user wants to play again or exit to the main menu.
8. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.6: Quadruple Line Clear

1. The game begins with a random tetrominos falling.
2. The user creates a stack of tetrominos on the grid.
3. The user uses a tetrominos to clear four lines.
4. The game rewards the user 800 points times the level of the game.
5. The user gets a game over.
6. The program displays the user their score.
7. The program asks if the user wants to play again or exit to the main menu.
8. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.7: T-Spin Single Line Clear

1. The game begins with a random tetrominos falling.
2. The user creates a stack of tetrominos on the grid in a certain way where putting a T piece will clear one line.
3. The user uses a T piece to clear one line.
4. The game rewards the user 800 points plus the level of the game.
5. The user gets a game over.
6. The program displays the user their score.
7. The program asks if the user wants to play again or exit to the main menu.
8. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.8: T-Spin Double Line Clear

1. The game begins with a random tetrominos falling.
2. The user creates a stack of tetrominos on the grid in a certain way where putting a T piece will clear two lines.
3. The user uses a T piece to clear two lines.
4. The game rewards the user 1200 points plus the level of the game.
5. The user gets a game over.
6. The program displays the user their score.
7. The program asks if the user wants to play again or exit to the main menu.
8. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.9: T-Spin Triple Line Clear

1. The game begins with a random tetrominos falling.
2. The user creates a stack of tetrominos on the grid in a certain way where putting a T piece will clear three lines.
3. The user uses a T piece to clear three lines.
4. The game rewards the user 1600 points plus the level of the game.
5. The user gets a game over.
6. The program displays the user their score.
7. The program asks if the user wants to play again or exit to the main menu.
8. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.10: 7-Bag

1. The game begins with a random tetrominos falling.
2. The user places 7 unique tetrominos.
3. The program generates 7 more unique tetrominos.
4. The user places 7 more tetrominos.
5. The user gets a game over.
6. The program displays the user their score.
7. The program asks if the user wants to play again or exit to the main menu.
8. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.11: Rotate Piece

1. The game begins with a random tetrominos falling.
2. The user rotates a piece and places it down.
3. This process is done over the 7 bag cycle.
4. The user gets a game over.
5. The program displays the user their score.
6. The program asks if the user wants to play again or exit to the main menu.
7. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.12: Level Scaling

1. The game begins with the user at level 1 with a random tetrominos falling and the piece speed is at 0.01667G.
2. The user clears 10 lines and advances to level 2 with the piece's speed of 0.021017G.
3. The user clears 10 more lines and advances to the next level and the piece speed is calculated based on the following formula: $(0.8 - ((\text{level} - 1) * 0.007))^{\text{Level} - 1}$.
4. The user gets a game over.
5. The program displays the user their score.
6. The program asks if the user wants to play again or exit to the main menu.
7. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 2.13: Game Pause

1. The game begins with a random tetrominos falling.
2. The user pauses the game (with P).
3. The game state is stopped
4. The user unpauses the game.
5. The game counts down from 3.
6. The game state is resumed.
7. The user gets a game over.
8. The program displays the user their score.
9. The program asks if the user wants to play again or exit to the main menu.
10. END

Next Use Case: Mode selection, high score page, or exit program.

3. Score Attack Mode

Description: User selects to play the Score Attack Mode from the main menu.

Pre-condition: User selects to play the Score Attack Mode from the main menu.

Post-condition: User gets a game over and views their score and can choose to play again or exit to the main menu.

Use Case 3.1: 3 Minute Game (Success)

1. The user chooses a 3 minute game and the game starts and random tetrominos fall.
2. The user goes through the 3 minutes without getting a game over.
3. The program displays the user their score.
4. The program asks if the user wants to play again or exit to the main menu.
5. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 3.2: 3 Minute Game (Failure)

1. The user chooses a 3 minute game and the game starts and random tetrominos fall.
2. The user gets a game over before the 3 minutes.
3. The program displays the user their score.
4. The program asks if the user wants to play again or exit to the main menu.
5. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 3.3: 5 Minute Game (Success)

1. The user chooses a 5 minute game and the game starts and random tetrominos fall.
2. The user goes through the 5 minutes without getting a game over.
3. The program displays the user their score.
4. The program asks if the user wants to play again or exit to the main menu.
5. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 3.4: 5 Minute Game (Failure)

1. The user chooses a 5 minute game and the game starts and random tetrominos fall.
2. The user gets a game over before the 5 minutes.
3. The program displays the user their score.
4. The program asks if the user wants to play again or exit to the main menu.
5. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 3.5: 10 Minute Game (Success)

1. The user chooses a 10 minute game and the game starts and random tetrominos fall.
2. The user goes through the 10 minutes without getting a game over.
3. The program displays the user their score.
4. The program asks if the user wants to play again or exit to the main menu.
5. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 3.6: 10 Minute Game (Failure)

1. The user chooses a 10 minute game and the game starts and random tetrominos fall.
2. The user gets a game over before the 10 minutes.
3. The program displays the user their score.
4. The program asks if the user wants to play again or exit to the main menu.
5. END

Next Use Case: Mode selection, high score page, or exit program.

4. Line Clearing Challenge

Description: User selects to play the Line Clearing Challenge from the main menu.

Pre-condition: User selects to play the Line Clearing Challenge from the main menu.

Post-condition: User gets a game over and views their score and can choose to play again or exit to the main menu.

Use Case 4.1: Garbage blocks generator

1. The game begins with a random tetrominos falling.
2. The game generates a row of garbage blocks but one column is empty.
3. The user uses the tetromino to clear the line
4. The game generates another row of garbage blocks but one column is empty (but not on the game column as the previous one)
5. The user gets a game over.
6. The program displays the user their score.
7. The program asks if the user wants to play again or exit to the main menu.
8. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 4.2: Level Scaling

1. The game begins with a random tetrominos falling.
2. The game generates a row of garbage blocks but one column is empty.
3. The user uses the tetromino to clear the line
4. The game generates another row of garbage blocks but one column is empty (but not on the game column as the previous one)
5. When the user gets to level 2 after clearing 10 lines, the blocks fall and garbage blocks generate faster.
6. The user gets a game over.
7. The program displays the user their score.
8. The program asks if the user wants to play again or exit to the main menu.
9. END

Next Use Case: Mode selection, high score page, or exit program.

5. VS AI Mode

Description: User selects to play the Versus AI Mode from the main menu.

Pre-condition: User selects to play the Versus AI Mode from the main menu.

Post-condition: User gets a game over and views their score and can choose to play again or exit to the main menu.

Use Case 5.1: Marathon Mode (Player Wins)

1. The user selects to play the marathon mode against the AI.
2. The game begins with a random tetrominos falling.
3. The AI gets a game over.
4. The program displays the user winning the game and their score.
5. The program asks if the user wants to play again or exit to the main menu.
6. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 5.2: Marathon Mode (AI Wins)

1. The user selects to play the marathon mode against the AI.
2. The game begins with a random tetrominos falling.
3. The user gets a game over.
4. The program displays the user losing the game and their score.
5. The program asks if the user wants to play again or exit to the main menu.
6. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 5.3: Score Attack Mode (Player Wins)

1. The user selects to play the score attack mode against the AI.
2. The game begins with a random tetrominos falling.
3. The game has a set time limit (5 minutes).
4. The program compares the player and AI's score and determines the player has a greater score.
5. The program displays the user winning the game and their score.
6. The program asks if the user wants to play again or exit to the main menu.
7. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 5.4: Score Attack Mode (AI Wins)

1. The user selects to play the score attack mode against the AI.
2. The game begins with a random tetrominos falling.
3. The game has a set time limit (5 minutes).
4. The program compares the player and AI's score and determines the AI has a greater score.
5. The program displays the AI winning the game and their score.
6. The program asks if the user wants to play again or exit to the main menu.
7. END

Next Use Case: Mode selection, high score page, or exit program.

6. Custom Mode

Description: User selects to play the Custom Mode from the main menu.

Pre-condition: User selects to play the Custom Mode from the main menu.

Post-condition: User gets a game over and views their score and can choose to play again or exit to the main menu.

Use Case 6.1: User sets a 5x10 Grid (Least Grid Size)

1. The user selects to play with a 5x10 grid.
2. The user starts the game and the game begins with a random tetrominos falling.
3. The user gets a game over.
4. The program displays the user their score.
5. The program asks if the user wants to play again or exit to the main menu.
6. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 6.2: User sets a 20x40 Grid (Most Grid Size)

1. The user selects to play with a 20x40 grid
2. The user starts the game and the game begins with a random tetrominos falling.
3. The user gets a game over.
4. The program displays the user their score.
5. The program asks if the user wants to play again or exit to the main menu.
6. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 6.3: User Disables a Tetrominos

1. The user selects the I, J, and L pieces to be removed.
2. The user starts the game and the game begins with a random tetrominos falling (excluding the ones the user removed).
3. The bag of pieces becomes 4.
4. The user gets a game over.
5. The program displays the user their score.
6. The program asks if the user wants to play again or exit to the main menu.
7. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 6.4: User Creates a Custom Tetris Piece

1. The user creates a 3x3 tetromino.
2. The user starts the game and the game begins with a random tetrominos falling.
3. The bag of pieces becomes 8.
4. The user gets a game over.
5. The program displays the user their score.
6. The program asks if the user wants to play again or exit to the main menu.
7. END

Next Use Case: Mode selection, high score page, or exit program.

7. High Score Page

Description: User selects to view their high scores from the main menu.

Pre-condition: User selects to view the high scores from the main menu.

Post-condition: User exits the high score page and goes back to the main menu.

Use Case 7.1: User Visits High Score Page With No Data

1. User visits the high score page
2. The program says there is no data in here.
3. The user exits the page.
4. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 7.2: User Imports High Score Data Into The Page (Successful)

1. User visits the high score page
2. The program says there is no data in here.
3. The user imports a text file with the high scores.
4. The program was successfully able to import the file.
5. The user looks at the high scores.
6. The user exits the page.
7. END

Next Use Case: Mode selection, high score page, or exit program.

Use Case 7.3: User Imports High Score Data Into The Page (Failure)

1. User visits the high score page
2. The program says there is no data in here.
3. The user imports a text file with the high scores.
4. The program was unsuccessful at importing the file.
5. The user exits the page.
6. END

Next Use Case: Mode selection, high score page, or exit program.

Sources:

Scoring System: <https://tetris.wiki/Scoring>

Game Difficulty: <https://tetris.wiki/Marathon>